Kaloyan (Kal) Draganov

Boston / Sofia (EU citizen) | kaloyan@draganov.com | +1 857-361-2905 | January – June 2027 Full-time Internship/Co-op Availability
https://linkedin.com/in/kaloyan-draganov-081567155/ | https://kkdraganov.github.io | https://github.com/kkdraganov

EDUCATION

Northeastern University, Boston, US **GPA 3.9/4.0 (Dean's List)**
Third year candidate for Bachelor of Science in Computer Engineering and Computer Science
Activities: Northeastern Mars Rover Team, AerospaceNU Avionics, Wireless Club, Cooking Club
Notable Coursework: Embedded + Mobile Robotics, Computer Systems, Electronics, OOD, Networks, Algorithms, Circuits & Signals

TECHNICAL SKILLS

Programming: Python, C, C++, Java, C#/.NET, SQL, PyTorch, MATLAB, Git
Embedded/Robotics: ROS 2, STM32, FreeRTOS, Altium, FPGA, Linux/Ubuntu, I²C, SPI, UART, Oscilloscope, Logic Analyzer, SMD soldering
Languages: Native Bulgarian, Native English, Basic German

WORK EXPERIENCE

Re:Build Fikst **January 2026 – June 2026**
Embedded Systems Engineering Intern Wilmington MA, US

- Developed embedded firmware in C for STM32/FreeRTOS, implementing device control and system logic alongside TouchGFX GUI workflows and support for new PCB revisions and peripheral variants
- Performed hardware bring-up for the device main PCB using an oscilloscope and logic analyzer, developing I²C drivers for the onboard EEPROM and voltage-supervisor ICs and identifying a PCB layout issue that prompted a board respin
- Owned two projects developing production systems to automate procedures in medical device manufacturing
  - Cassette Flasher: Designed and built a Python-based standalone EEPROM programming station on Ubuntu using FT232H with multi-threaded hardware control and systemd deployment; used for every manufactured consumable
  - Main PCB End-of-Line QC: Led development of a fault-tolerant automated PCB test station implementing custom Windows/Linux DAQ drivers with hardware timeout handling and safe shutdown; validates every production PCB
- Unified two core components of the company's embedded development infrastructure by implementing support for its proprietary serial protocol within the embedded prototyping framework, resolving a longstanding interoperability gap

Uru Intelligence, LLC **April 2025 – September 2025**
Founder & CTO Wellesley MA, US

- Architected technical stack from the ground up: React/FastAPI/Supabase app in Docker Compose hosted on Elestio
- Engineered an MCP-driven integration layer to extend standard LLMs to utilize third-party apps (e.g. Gmail, Calendar, Teams)
- Designed AI pipelines leveraging n8n and OpenAI agents for custom indexing with RAG vector databases in Pinecone

Dynamo Software **January 2022 – June 2024**
Developer Watertown MA, US

- Independently scoped, designed, and delivered core product features in Python, SQL, and C# after collaborating with clients to define key use cases and translate business needs into technical specifications
- Architected data pipelines and backend services integrating SQL Server and RESTful APIs for automated data processing
- Designed a "signature parser" microservice in C#, creating custom pattern recognition and similarity algorithms, achieving 90% accuracy in extracting contact data from client emails and reducing manual processing time by 95%

TECHNICAL CLUBS & PROJECTS

Northeastern Mars Rover Team **March 2025 – Present**
Autonomous Software Lead *Placed 3rd URC '25, 6th URC '26, 2nd CIRC '26*

- Implemented EKF sensor fusion in ROS 2 on an NVIDIA Jetson Orin running embedded Linux, fusing GNSS and frame-transformed ZED IMU odometry; developed noisy-sensor simulation and ground-truth visualization for validation
- Developed a holonomic MPPI local planner for obstacle avoidance with stochastic control sampling and tuned cost functions
- Migrated drivetrain motor control from PWM to CAN, integrating Kraken X44 motors on a 1 MBd SocketCAN 2.0 bus

AerospaceNU Avionics **September 2025 – December 2025**
Embedded Engineer

- Developed flight control and telemetry software in PlatformIO C++ on an ARM Cortex-M0 processor, implementing real-time control logic for the flight computer, telemetry streams, engine ignition, and launch sequencing
- Integrated custom PCBs with onboard sensors and peripherals over SPI, I²C, and UART, ensuring synchronized operation

INTERESTS

Robotics, Travel, Arduino, Beekeeping, 3D Printing, Computer Vision, Soccer, Rock Climbing, Skiing, Sim Racing, Reading